

relative periods into $[-T, -K)$ and $(L, T]$ and estimate a “binned” specification

$$Y_{i,t} = \alpha_i + \lambda_t + \beta \cdot \sum_{\ell < -K} D_{i,t}^\ell + \sum_{\ell = -K}^{-2} \mu_\ell D_{i,t}^\ell + \sum_{\ell=0}^L \mu_\ell D_{i,t}^\ell + \gamma \cdot \sum_{\ell > L} D_{i,t}^\ell + v_{i,t} \quad (8)$$

without excluding any distant relative periods so that $g^{excl} = \{-1\}$. For “trimming”, researchers trim their panel to be balanced in relative periods.

Neither “binning” nor “trimming” resolves the issues of contamination discussed below (i.e. the possibility that treatment effects from other periods affect the estimate for a given μ_g). We show the contamination issue for the general specification (4) encompasses both practices. For a given coefficient in the dynamic specification, “trimming” does however mechanically remove any treatment effects from the relative periods “trimmed” from the specification. For the static specification put forth in Borusyak and Jaravel (2017), they noted that “trimming” also does not resolve the contamination issue they identified with the static specification.

3.2 Interpreting the coefficients under no assumptions

First we show that without any assumptions, we can write μ_g as a linear combination of differences in trends.

Proposition 1. *The population regression coefficient on relative period bin g is a linear combination of differences in trends from its own relative period $\ell \in g$, from relative periods $\ell \in g'$ belonging to other bins $g' \neq g$ but included in the specification, and from relative periods excluded from the specification $\ell \in g^{excl}$:*

$$\mu_g = \sum_{\ell \in g} \sum_e \omega_{e,\ell}^g \left(E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i = e] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] \right) \quad (9)$$

$$+ \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell \in g'} \sum_e \omega_{e,\ell}^g \left(E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i = e] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] \right) \quad (10)$$

$$+ \sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g \left(E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i = e] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] \right). \quad (11)$$

We use the superscript g to associate the weight $\omega_{e,\ell}^g$ with the coefficient μ_g . The weight $\omega_{e,\ell}^g$ is equal to the population regression coefficient on $\mathbf{1}\{t - E_i \in g\}$ from regressing $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ on all bin indicators $\{\mathbf{1}\{t - E_i \in g\}\}_{g \in \mathcal{G}}$ included in the specification (4) and two-way fixed effects.

The above proposition is a direct result of regression mechanics. We provide an intuitive derivation for

the closed-form expressions for the weights using the classical “omitted variables bias formula” in Section 3.6. We defer the formal derivation to the Appendix. Here we mention the following four properties of the weights $\omega_{e,\ell}^g$.

- For relative periods of μ_g ’s own bin i.e. $\ell \in g$, their associated weights as displayed in (9) sum to one $\sum_{\ell \in g} \sum_e \omega_{e,\ell}^g = 1$.
- For relative periods belonging to some other bin included in (4) i.e. $\ell \in g'$ for $g' \neq g$ and $g' \in \mathcal{G}$, their associated weights as displayed in (10) sum to zero $\sum_{\ell \in g'} \sum_e \omega_{e,\ell}^g = 0$ for each bin g' .
- For relative periods not included in $\mathcal{G}$, their associated weights as displayed in (11) sum to negative one $\sum_{\ell \in g^{excl}} \sum_e \omega_{e,\ell}^g = -1$.
- If there are never-treated units i.e. $\infty \in \text{supp}(E_i)$, we have $\omega_{\infty,\ell}^g = 0$ for all g and ℓ .

We can easily estimate the weights $\omega_{e,\ell}^g$ for any given specification of $\mathcal{G}$ using the following auxiliary regression:

$$D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \omega_{e,\ell}^g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (12)$$

which regresses $D_{i,t}^\ell \cdot \mathbf{1}\{E_i = e\}$ on all bin indicators included in regression (4) and two-way fixed effects.

All of the above properties can be extended to a case where covariates are added to regression (4) by partialling out the covariates before proceeding. In other words, the terms in parentheses in (9), (10) and (11) would be replaced by terms for which the covariates are partialled out. The weights can be estimated by controlling for covariates in regression (12) the same way as they are controlled for in the original regression.

However, covariates complicate the interpretations of μ_g in terms of $CATT_{e,\ell}$ as we describe below in Proposition 2-4. Depending on how covariates are controlled for in regression (4), we may need an additional assumption that the counterfactual trends are linear in the time-varying covariates $X_{i,t}$. We leave a full investigation of the introduction of covariates for future work.

3.3 Interpreting the coefficients under parallel trends assumption only

Proposition 2. *Under Assumption 1 (parallel trends) only, the population regression coefficient on the indicator for relative period bin g is a linear combination of $CATT_{e,\ell \in g}$ as well as $CATT_{e,\ell'}$ from other*

relative periods $\ell' \notin g$, with the same weights stated in Proposition 1:

$$\mu_g = \sum_{\ell \in g} \sum_e \omega_{e,\ell}^g CATT_{e,\ell} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g'} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (13)$$

Under Assumption 1, the terms in Proposition 1 reduce to a linear combination of the causally interpretable building blocks $CATT_{e,\ell}$ as follows:

$$E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] = CATT_{e,\ell} + \underbrace{E[Y_{i,t}^\infty - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty]}_{=0} \quad (14)$$

for $t = e + \ell$. However, two issues for interpretability remain. First, the coefficient μ_g can be written as an average of not only $CATT_{e,\ell}$ from own periods $\ell \in g$, but also $CATT_{e,\ell'}$ from other periods. Second, the weights are still non-linear functions of the distribution of cohorts, same as those in (12), and they are not restricted to lie in $[0, 1]$.

The properties of these weights as described following Proposition 1 imply that contamination from other periods wanes once we impose restrictions on treatment effects. In the next two subsections we illustrate how that can happen.

3.4 Interpreting the coefficients under parallel trends and no anticipation assumptions

Proposition 3. *If Assumption 1 (parallel trends) holds and Assumption 2 (no anticipatory behavior in all periods before the initial treatment) holds, the population regression coefficient μ_g is a linear combination of post-treatment $CATT_{e,\ell'}$ for all $\ell' \geq 0$, with the same weights stated in Proposition 1:*

$$\mu_g = \sum_{\ell' \in g, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g', \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (15)$$

Once we restrict pre-treatment $CATT_{e,\ell \leq 0}$ to be zero under the no anticipatory behavior assumption, the expression for μ_g simplifies as terms involving $CATT_{e,\ell \leq 0}$ drop out. However, the second term in the expression for μ_g remains unless we further impose treatment effect homogeneity for its summands to cancel out each other. Thus, μ_g may be non-zero for pre-treatment periods even if parallel trends holds.

This result immediately implies a shortcoming of using pre-treatment coefficients (i.e. μ_g where g contains only leads to the treatment $\ell < 0$) to test for pretrends. Under the no anticipatory behavior assumption,

cohort-specific treatment effects prior to treatment are all zero: $CATT_{e,\ell} = 0$ for all $\ell < 0$. Therefore, any linear combination of these $CATT_{e,\ell}$ is also zero. However, μ_g is a function of post-treatment $CATT_{e,\ell' \geq 0}$ as well, even when g only contains elements with $\ell < 0$. We revisit this implication in greater depth in Section 3.7. Callaway and Sant’Anna (2020a) provides alternative tests for pretrends that do not suffer from this drawback.

3.4.1 Sources of treatment effect heterogeneity

Since treatment effects heterogeneity violates Assumption 3 and can alter how we interpret μ_g , it is important to think through when different cohorts likely experience different paths of treatment effect. Such heterogeneity could arise for many reasons. For example, cohorts may differ in their covariates, which affect how they respond to treatment. We will explore a concrete example in our application: if treatment effects differ with age, and there is variation in age across units first treated at different times, we will have heterogeneous effects (see Section 5 for details). After controlling for covariates, cohorts may still vary in their responses to the treatment if units select their initial treatment timing based on treatment effects. This source of heterogeneity is still compatible with our parallel trends assumption, which only rules out selection in the initial treatment timing based on the evolution of the baseline outcome. In addition to these two sources of heterogeneity, treatment effects may vary across cohorts due to calendar time-varying effects (e.g. macroeconomic conditions could govern the effects on labor market outcomes across cohorts).

3.5 Interpreting the coefficients under parallel trends and treatment effect homogeneity

Proposition 4. *If Assumption 1 (parallel trends) holds and Assumption 3 (treatment effect homogeneity) holds, then $CATT_{e,\ell} = ATT_\ell$ is constant across e for a given ℓ , and the population regression coefficient μ_g is equal to a linear combination of $ATT_{\ell \in g}$, as well as $ATT_{\ell' \notin g}$ from other relative periods:*

$$\mu_g = \sum_{\ell \in g} \omega_\ell^g ATT_\ell + \sum_{g' \neq g} \sum_{\ell' \in g'} \omega_{\ell'}^g ATT_{\ell'} + \sum_{\ell' \in g^{excl}} \omega_{\ell'}^g ATT_{\ell'} \quad (16)$$